

Alonso Bardales

Python AI Engineer | LLM Applications, Agent Pipelines, RAG

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SUMMARY

Python developer focused on LLM application development, autonomous agent pipelines, and RAG. Builds end-to-end AI systems with the Anthropic Messages API (tool use and structured outputs), ChromaDB vector search, and production concerns such as cost control, OAuth integration, and crash-safe pipelines. Applied AI student at Miami Dade College who builds and ships tools end to end.

SKILLS

AI and LLM Engineering: LLM application development, agent tool use and function calling, Anthropic Messages API, structured outputs, RAG pipelines, vector databases (ChromaDB), embeddings and retrieval ranking, prompt engineering, NLP

Python and Backend: Python, Flask, FastAPI, httpx, typer, pydantic, REST API design and integration, SQL (SQLite, PostgreSQL)

Cloud and Tooling: cloud deployment (Render), GitHub Actions, OAuth2, JWT, AES-GCM, Git

Frontend and Mobile: React, TypeScript, Swift, SwiftUI

EXPERIENCE

MicroFinance Pro, Miami, FL

Sep 2025 to Present

- Designed and integrated multi-agent AI workflows with Microsoft Azure AI Foundry to automate business processes and deliver multilingual translation features.

EPIC Innovation Challenge, Miami Tech Works / Riipen / U.S. Chamber of Commerce

Aug 2025 (6 weeks)

- Worked on a cross-functional team to design and build a Learning Management System that teaches AI literacy and certifies users on completion.

PROJECTS

- vibeMusicians (Python 3.11, Anthropic Messages API, httpx, typer, pydantic, SQLite):** Six-stage autonomous AI agent pipeline that researches music trends, invents an artist persona, writes a song, generates audio and cover art, and publishes to SoundCloud; ran unattended on a GitHub Actions cron and published 26 original tracks over roughly two weeks. Uses the web search tool with bounded usage for cost control and JSON schema structured outputs, per-stage SQLite checkpointing for crash-safe resume, SoundCloud OAuth 2.1 single-use rotating refresh tokens persisted across ephemeral CI runs with an Actions concurrency guard, and a provider-agnostic client with defensive parsing across three vendor APIs.
- ai-assistant (Python, Anthropic Messages API, ChromaDB, pdfplumber, Whisper):** Local Claude agent with a twelve-tool function-calling loop. RAG pipeline using 800-word chunks with 100-word overlap, a ChromaDB persistent vector store with ONNX all-MiniLM-L6-v2 embeddings, md5-keyed dedup on re-indexing, and relevance-ranked retrieval. Integrates OAuth2 for Google Calendar and Gmail with token refresh, pdfplumber for PDF question answering, and local Whisper for speech to text.
- Tutorio, AI English and Math Tutor (Flask, Render, SQLite):** Deployed an LLM tutoring platform with layered rate limiting in SQLite, API-key auditing, COPPA age-gating, and system-prompt guardrails against tutor drift.
- Priority Task Manager, iOS (Swift, SwiftUI):** Task management app built and shipped solo to the App Store with full ownership from design through submission and review; iCloud sync via NSUbiquitousKeyValueStore with conflict resolution, and a natural-language parser that turns free-form input into structured plans. Also shipped Quintave Finance Tracker to the App Store.

EDUCATION

Miami Dade College, Miami, FL, Bachelor of Science in Applied Artificial Intelligence

Phi Theta Kappa Honor Society, Active Member; Scholastic Art and Writing Awards, Gold Key (2022)

Expected graduation: December 2027